

DATA SCIENCE & AI MASTERS 2026

150-DAY DEEP-MASTERY STUDY PLAN - 8 HOURS/DAY - 18 PHASES

Replanned from the uploaded 120-day / 14-phase master plan

Plan element	Updated target
Course basis	25 sections 632 lectures 12 quizzes 4 role-plays 115h 56m
Study commitment	8 focused study hours per day
Time available	150-day deep-mastery window
Phase structure	Exactly 18 phases
Day 1-4	Preserved from the previous Phase 1 without content changes
Learning method	Learn -> reproduce -> independent practice -> review -> build -> explain
Primary outcome	Job-ready evidence: skills, projects, deployment, interview readiness

What changed: The same source curriculum is spread across 150 days instead of 120, and the roadmap is split into 18 clearer phases. Additional time is used for reproduction, independent coding, review, error analysis, interview practice and project consolidation.

8-hour daily framework: 120 min new concepts/lectures + 120 min code-along/reproduction + 120 min independent coding + 60 min notes/active recall + 60 min LeetCode/HackerRank/quiz/error log = 8 focused study hours. Use breaks between major blocks.

Planning principle: The 150-day extension does not require adding unrelated material. It gives high-value topics more repetition, independent implementation, debugging, interview practice and project depth.

18-PHASE MASTER ROADMAP

#	Phase	Source	Days	Primary objective
1	Programming Foundations	Section 02	1-4	Core Python setup, variables, types, operators, collections, control flow and functions.
2	Python Data Stack & Foundation Practice	Section 02 + mastery extension	5-8	Turn Python fundamentals into repeatable data-science workflows.
3	Statistical Foundations	Section 03	9-15	Build practical descriptive and inferential statistics for data-science interviews and modeling.
4	Data Analysis & Exploration	Section 04	16-22	Develop a complete EDA process from sourcing and cleaning through reporting.
5	Database & Query Skills	Section 05	23-29	Become independently productive with SQL and connect SQL workflows to Python.
6	Machine Learning - Foundations & Regression	Section 06	30-38	Understand the ML workflow and become strong in regression, evaluation and regularization.
7	Machine Learning - Classification & Ensembles	Section 06	39-48	Master core classification methods, evaluation and ensemble reasoning.
8	Machine Learning - Clustering & Feature Engineering	Section 06	49-57	Build unsupervised-learning intuition and practical feature-engineering skills.
9	Machine Learning - HPO, Projects & Mastery	Section 06	58-64	Turn ML knowledge into reproducible model-selection and optimization capability.
10	Time Series Forecasting	Section 07	65-72	Understand time-series structure and build ARIMA/Prophet forecasting workflows.
11	Deep Learning & Neural Networks	Section 08	73-81	Build strong conceptual and practical understanding of ANN, CNN, RNN/LSTM and transfer learning.
12	Natural Language Processing	Section 09	82-91	Develop practical classical NLP, embeddings and sequence-modeling capability.
13	Transformers & Large Language Models	Section 10	92-102	Understand transformer architecture, BERT/GPT families, prompting and LLM fundamentals.
14	RAG, LangChain & LLM Applications	Section 10	103-112	Build retrieval-augmented systems and understand the end-to-end LLM application stack.
15	Advanced Generative AI & RAG Evaluation	Sections 11-12	113-121	Understand fine-tuning choices and rigorously evaluate RAG/GenAI systems.
16	Agentic AI, LangGraph & Automation	Sections 13-15	122-132	Build tool-using agents, understand agentic workflows and automate AI tasks.
17	Production Engineering, Data Engineering & Cloud	Sections 16-17, 23	133-141	Serve, deploy and operationalize ML/GenAI systems with foundational data-engineering knowledge.
18	Industry Capstones, Interview Mastery & Final Consolidation	Sections 18-25 + mastery	142-150	Convert course knowledge into demonstrable projects, interview readiness and a final portfolio.

Day 1-4 preservation: The first four daily focus items remain as in the existing plan; the 150-day reallocation begins from Day 5 onward.

PHASE 1 - Programming Foundations

Source: Section 02 **Study days:** 1-4

Phase objective: Core Python setup, variables, types, operators, collections, control flow and functions.

Preservation note: Days 1-4 are intentionally unchanged from the earlier plan.

Day	Focus	8-hour execution target
1	Python setup + fundamentals	Installation, Colab, variables, types, operators. Code every example.
2	Python collections	Lists, tuples, sets, dictionaries, indexing and slicing.
3	Control flow + functions	Loops, iterations, control structures, functions.
4	Reusable Python	Map/reduce/filter, file handling, OOP.

Phase deliverable: A clean Python notebook covering fundamentals plus a small reusable script.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 2 - Python Data Stack & Foundation Practice

Source: Section 02 + mastery extension **Study days:** 5-8

Phase objective: Turn Python fundamentals into repeatable data-science workflows.

Day	Focus	8-hour execution target
5	Python data stack	NumPy, Pandas, Matplotlib, Seaborn; build a small EDA script and complete the Python quiz.
6	Reproduce from memory	Rebuild key Python examples without the instructor; focus on clean syntax and debugging.
7	Data workflow practice	Combine Python + NumPy + Pandas + visualization on a fresh dataset.
8	Foundation checkpoint	Interview review, coding challenge set, error log, and independent mini-project.

Phase deliverable: One end-to-end Python data workflow and a Python interview cheat sheet.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 3 - Statistical Foundations

Source: Section 03 **Study days:** 9-15

Phase objective: Build practical descriptive and inferential statistics for data-science interviews and modeling.

Day	Focus	8-hour execution target
9	Statistics foundations	Types of data, descriptive vs inferential statistics, sampling, population vs sample.
10	Central tendency	Mean, median, mode; when each is informative.
11	Dispersion	Range, IQR, variance, standard deviation, mean deviation.
12	Probability	Addition rule, independent events, conditional probability, Bayes theorem.
13	Distributions	Uniform, binomial, Poisson, normal distribution, skewness, kurtosis, Z-scores.
14	Relationships	Covariance, correlation and interpretation.
15	Inference + review	p-values, t-test, z-test, chi-square, ANOVA, correlation test; problem set + quiz.

Phase deliverable: 1-page statistics cheat sheet, solved problem set, and Python implementations of core tests.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 4 - Data Analysis & Exploration

Source: Section 04 **Study days:** 16-22

Phase objective: Develop a complete EDA process from sourcing and cleaning through reporting.

Day	Focus	8-hour execution target
16	EDA process	DA/DS processes, EDA steps, data sourcing and cleaning.
17	Missing data	Missing-value theory and practical handling.
18	Scaling	Standardization and normalization; practical implementation.
19	Outliers	Outlier theory, detection and treatment; invalid data.
20	Analysis types	Univariate, bivariate and multivariate analysis; numerical analysis.
21	Feature transformation	Derived metrics, feature binning and feature encoding.
22	EDA mastery	Case study, EDA report, fresh dataset end-to-end, quiz/review.

Phase deliverable: A polished EDA report with documented cleaning and transformation decisions.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 5 - Database & Query Skills

Source: Section 05 **Study days:** 23-29

Phase objective: Become independently productive with SQL and connect SQL workflows to Python.

Day	Focus	8-hour execution target
23	SQL foundations	Architecture, installation, constraints, table basics, DDL, DQL, DML.
24	Joins	Inner/outer join logic and practical multi-table querying.
25	Functions	Aggregation, string and date-time functions.
26	Regex + nested queries	Regular expressions, subqueries and query composition.
27	Views + procedures	Views and stored procedures; understand when each helps.
28	Window functions	Ranking and analytical patterns using window functions.
29	SQL-Python mastery	SQL-Python connectivity, 15+ practice queries, quiz and review.

Phase deliverable: A query portfolio with joins, windows, subqueries, procedures and Python connectivity.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 6 - Machine Learning - Foundations & Regression

Source: Section 06 **Study days:** 30-38

Phase objective: Understand the ML workflow and become strong in regression, evaluation and regularization.

Day	Focus	8-hour execution target
30	ML foundations	Types of ML, use cases, features, train-test split, scaling and encoding.
31	Regression evaluation	Regression metrics, error thinking and workflow.
32	Linear regression I	Simple linear regression theory and practical implementation.
33	Linear regression II	Multiple linear regression, interpretation and diagnostics.
34	Polynomial regression	Nonlinear regression and polynomial feature behavior.
35	Bias-variance	Underfitting, overfitting and bias-variance trade-off.
36	Ridge + Lasso	Regularization concepts, comparison and practicals.
37	Independent regression	Build and compare regression models on a fresh dataset.
38	Regression checkpoint	Closed-book review, model explanation and targeted gap fixing.

Phase deliverable: Regression project, model-comparison notes and evaluation checklist.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 7 - Machine Learning - Classification & Ensembles

Source: Section 06 **Study days:** 39-48

Phase objective: Master core classification methods, evaluation and ensemble reasoning.

Day	Focus	8-hour execution target
39	Classification foundations	Log loss, confusion matrix, AUC-ROC and classification report.
40	kNN	Distance intuition, k selection and practical implementation.
41	Decision trees	Decision trees, entropy, Gini and visualization.
42	Random Forest	Ensemble intuition and practical implementation.
43	Naive Bayes	Probability-based classification intuition and use cases.
44	SVM	Margin intuition, kernels at a conceptual level and practicals.
45	Logistic regression	Decision boundary, probabilities and practical implementation.
46	Classification project	Build and evaluate a full classification solution.
47	Ensemble learning	Bagging, boosting, AdaBoost, Gradient Boosting and XGBoost.
48	Classification checkpoint	Error analysis, model comparison and quiz-style review.

Phase deliverable: Classification project plus a table explaining why one model was selected over others.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 8 - Machine Learning - Clustering & Feature Engineering

Source: Section 06 **Study days:** 49-57

Phase objective: Build unsupervised-learning intuition and practical feature-engineering skills.

Day	Focus	8-hour execution target
49	Clustering foundations	k-Means and the clustering objective.
50	Hierarchical + Mean Shift	Hierarchical clustering and Mean Shift; compare cluster behavior.
51	Cluster interpretation	Evaluate and interpret clusters in business/data-science terms.
52	Feature engineering I	RFE, SFS, Chi-Square and feature-selection reasoning.
53	Feature engineering II	PCA and LDA; interpretation and practical use.
54	kPCA + QDA	Kernel PCA and QDA; understand nonlinear boundaries/features.
55	Feature-engineering practice	Compare raw vs engineered feature pipelines.
56	Independent ML workflow	Use preprocessing, feature engineering and multiple model families.
57	Feature checkpoint	Closed-book review and interview questions on selection, dimensionality reduction and clustering.

Phase deliverable: Feature-engineering notebook with justified transformations and cluster interpretation.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 9 - Machine Learning - HPO, Projects & Mastery

Source: Section 06 **Study days:** 58-64

Phase objective: Turn ML knowledge into reproducible model-selection and optimization capability.

Day	Focus	8-hour execution target
58	Manual HPO	Hyperparameter reasoning and validation design.
59	GridSearch	Systematic search, validation and comparison.
60	RandomizedSearch	Random search and when it is preferable to exhaustive search.
61	Independent ML project I	One dataset through preprocessing, three model families and baseline comparison.
62	Independent ML project II	Feature engineering, tuning and error analysis.
63	Independent ML project III	Final model choice, documentation and explanation.
64	ML mastery checkpoint	Closed-book review, code from memory, quiz and gap fixing.

Phase deliverable: One complete ML project with model comparison, HPO record and final rationale.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 10 - Time Series Forecasting

Source: Section 07 **Study days:** 65-72

Phase objective: Understand time-series structure and build ARIMA/Prophet forecasting workflows.

Day	Focus	8-hour execution target
65	Time-series foundations	Components, decomposition and anomaly detection.
66	Stationarity	Stationarity testing, transformations and preprocessing.
67	AR/MA	Autoregression, moving average and the intuition behind ARIMA.
68	ACF/PACF	Use ACF/PACF to reason about p/d/q.
69	ARIMA practicals	Fit, tune, grid search and select a final model.
70	Prophet	Prophet theory and multivariate forecasting concepts.
71	Forecasting metrics	MSE, RMSE, MAPE and practical model comparison.
72	Forecasting mastery	Run an independent forecast and compare ARIMA vs Prophet.

Phase deliverable: Forecasting report with assumptions, metric comparison and model-selection reasoning.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 11 - Deep Learning & Neural Networks

Source: Section 08 **Study days:** 73-81

Phase objective: Build strong conceptual and practical understanding of ANN, CNN, RNN/LSTM and transfer learning.

Day	Focus	8-hour execution target
73	Deep learning foundations	Neuron, activation functions, forward pass, backpropagation and gradient descent.
74	ANN practicals	ANN implementation, optimization, debugging and intuition.
75	CNN fundamentals	Convolution, pooling, architecture, augmentation and batch/epoch concepts.
76	CNN practicals	Build CNN, inspect parameters, evaluate the image pipeline.
77	RNN	RNN basics, types and sequence modeling.
78	Gradient problems	Vanishing/exploding gradients and why recurrence is difficult.
79	LSTM	LSTM architecture, gates and practicals.
80	Transfer learning	VGG16, MobileNet and transfer-learning workflow.
81	DL project checkpoint	Retrain an image model on a new dataset; quiz and review.

Phase deliverable: Working image-model notebook plus architecture explanation.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 12 - Natural Language Processing

Source: Section 09 **Study days:** 82-91

Phase objective: Develop practical classical NLP, embeddings and sequence-modeling capability.

Day	Focus	8-hour execution target
82	NLP foundations	Intro, linguistics, challenges, case folding and cleanup.
83	Tokenization	Tokenization, stop-word handling and text preprocessing.
84	Classical representations	n-grams, vectorization and Bag of Words.
85	TF-IDF	TF-IDF theory and practical implementation.
86	Information extraction	POS tagging and NER; practical pipeline.
87	Word2Vec	CBOW, Skip-Gram, pretrained Word2Vec and embedding intuition.
88	GloVe + FastText	Compare embedding families and cosine similarity.
89	LSTM for NLP	Architecture, pointwise operations and forget/input/output gates.
90	GRU + BiLSTM	GRU gates, practicals and bidirectional LSTM.
91	NLP checkpoint	Build a text classification/similarity task; quiz and review.

Phase deliverable: NLP notebook comparing representation choices and a small sequence-model task.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 13 - Transformers & Large Language Models

Source: Section 10 **Study days:** 92-102

Phase objective: Understand transformer architecture, BERT/GPT families, prompting and LLM fundamentals.

Day	Focus	8-hour execution target
92	Transformer architecture	Transformer types, core architecture and self-attention.
93	Encoder	Encoder stack and contextual embeddings.
94	Decoder	Decoder architecture and autoregressive generation.
95	BERT	BERT configurations, masked language modeling and fine-tuning.
96	BERT family	RoBERTa, DistilBERT and ALBERT; compare design choices.
97	GPT architecture	Decoder-only architecture, masked multi-head attention and blocks.
98	GPT training	Training workflow and causal next-token prediction.
99	LLM foundations	Tokens, context windows and prompt fundamentals.
100	Prompt engineering	Prompt engineering, prompt tuning and prompt structures.
101	Transformer reasoning	Explain self-attention, encoder/decoder roles and model-family trade-offs.
102	LLM checkpoint	Closed-book architecture review and interview-style explanation practice.

Phase deliverable: Transformer architecture notes, comparison sheet and verbal explanation recording.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 14 - RAG, LangChain & LLM Applications

Source: Section 10 **Study days:** 103-112

Phase objective: Build retrieval-augmented systems and understand the end-to-end LLM application stack.

Day	Focus	8-hour execution target
103	RAG foundations	What/why, use cases, retrieval architecture and generation flow.
104	RAG ingestion	Document ingestion, chunking and retrieval concepts.
105	PDF RAG	Build a PDF RAG chatbot and understand each component.
106	LangChain	Framework concepts and application structure.
107	Prompting in RAG	Few-shot and chain-of-thought concepts; prompt design.
108	Vector databases	Vector DB vs vector index; similarity search.
109	Pinecone practicals	Implement vector storage/retrieval and inspect retrieval behavior.
110	LSH + retrieval quality	Locality-sensitive hashing and retrieval experiments.
111	Ollama	Setup, model testing and Python implementation.
112	Local RAG + APIs	Build/debug local RAG; understand API-backed workflows.

Phase deliverable: Independent PDF RAG chatbot, architecture diagram and retrieval error log.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 15 - Advanced Generative AI & RAG Evaluation

Source: Sections 11-12 **Study days:** 113-121

Phase objective: Understand fine-tuning choices and rigorously evaluate RAG/GenAI systems.

Day	Focus	8-hour execution target
113	RAG vs fine-tuning	Decision framework and use cases.
114	PEFT	Parameter-efficient fine-tuning concepts.
115	Quantization	Quantization purpose and trade-offs.
116	LoRA	Low-rank updates and adapter intuition.
117	qLoRA	Quantized low-rank adapters and workflow.
118	N-gram metrics	BLEU, ROUGE and METEOR.
119	Intrinsic metrics	Perplexity and what it measures.
120	Model-based metrics	BERTScore and BARTScore.
121	RAGAS + evaluation project	RAGAS framework, practical evaluation, failure analysis and quiz.

Phase deliverable: Evaluation report for your RAG system with multiple metrics and failure analysis.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 16 - Agentic AI, LangGraph & Automation

Source: Sections 13-15 **Study days:** 122-132

Phase objective: Build tool-using agents, understand agentic workflows and automate AI tasks.

Day	Focus	8-hour execution target
122	Agentic AI foundations	Agents, examples, tools and workflow patterns.
123	Built-in tools	LangChain built-in tools and tool integration.
124	Custom tools	Tool Decorator, StructuredTool and BaseTool.
125	Toolkits	Toolkit composition and tool selection.
126	Tool calling	Calling, execution loops and failure handling.
127	Agent mini-project	Currency-conversion tool and travel-planning agent.
128	Agent practicals	Reasoning, planning, autonomy and code walkthrough.
129	LangGraph	Stateful workflows and agentic workflow components.
130	LangGraph project	Build and debug an agentic workflow.
131	n8n foundations	Templates, nodes, workflow design and tool integration.
132	n8n AI automation	Build an AI-powered news summarizer and connect an agent tool.

Phase deliverable: One working tool-calling agent, one LangGraph workflow and one n8n AI automation.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 17 - Production Engineering, Data Engineering & Cloud

Source: Sections 16-17, 23 **Study days:** 133-141

Phase objective: Serve, deploy and operationalize ML/GenAI systems with foundational data-engineering knowledge.

Day	Focus	8-hour execution target
133	Deployment fundamentals	Deployment architecture and Flask basics.
134	ML API	Build a Flask model-serving application.
135	AWS fundamentals	EC2, networking concepts and deployment sequence.
136	AWS deployment	Deploy a model independently to EC2.
137	Data engineering	ETL, data warehouse, data marts and data lake.
138	GenAI deployment	Model serving, directory/PDF-based systems and Streamlit.
139	GenAI on AWS	Deploy a chatbot/application to EC2.
140	Production hardening	Document architecture, inputs/outputs, errors, dependencies and deployment steps.
141	Deployment checkpoint	Re-deploy one project end-to-end and verify it independently.

Phase deliverable: One deployed ML/GenAI application with architecture and deployment documentation.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

PHASE 18 - Industry Capstones, Interview Mastery & Final Consolidation

Source: Sections 18-25 + mastery **Study days:** 142-150

Phase objective: Convert course knowledge into demonstrable projects, interview readiness and a final portfolio.

Day	Focus	8-hour execution target
142	ML capstone	Cancer Risk Prediction: discovery, EDA, modeling and XGBoost/Optuna.
143	DL capstone	Solar Panel Defect Classification: CNN + transfer learning + deployment.
144	DL capstone	Elephant Species Classification: baseline + transfer learning + deployment.
145	GenAI capstones	ChatScholar, Research RAG, claims processing and multi-PDF RAG architecture patterns.
146	GenAI capstones	AI Career Coach, Sustainability Chatbot, Text-to-SQL and AWS Bedrock RAG.
147	Agentic + portfolio	Resume Genie + Dental Appointment System; choose 2-4 flagship projects.
148	Interview mastery	ML, DL and GenAI interview preparation; record answers and explain architectures.
149	Mock assessment	Six-to-eight-hour no-tutorial challenge: build, test, explain and document.
150	Final mastery	Client role-play, capability map, portfolio, resume/LinkedIn and next-role plan.

Phase deliverable: 2-4 flagship projects, recorded interview answers, client-role-play notes and updated portfolio.

8-hour rule: Do not spend the whole day watching. Reproduce the examples, code independently, review errors, and finish with active recall or an interview explanation.

FINAL 150-DAY MASTERY CHECKLIST

Area	End-state target
Python / NumPy / Pandas	Independent implementation and debugging
Statistics	Strong conceptual + practical understanding
EDA / SQL	Independent analysis workflow
Machine Learning	Strong model selection, evaluation, feature engineering and HPO
Time Series	Working forecasting capability
Deep Learning	Working ANN/CNN/RNN/LSTM/transfer-learning capability
NLP	Working classical NLP + sequence modeling capability
Transformers / LLMs	Strong conceptual + practical understanding
RAG / Evaluation	Strong practical build + rigorous evaluation
Agentic AI / LangGraph / n8n	Working tool-calling and automation capability
Deployment / AWS	Working deployed application
Portfolio	2-4 flagship, documented projects
Interview / consulting	Ready for technical + client conversations

Completion standard: Build without the instructor, explain without notes, debug when a system fails, and show an employer concrete project evidence. This retains the mastery standard of the source plan.

150-day target: The extended schedule is designed to make your learning durable, not merely longer. Use the extra days for repeated retrieval, independent builds, interview practice and polished portfolio evidence.